RETAIL BILLING WITH MULTIPLE DISCOUNT RULE

Python Activity-Based Learning (ABL) Assignment — Sample Implementation

1. Problem Statement

Develop a Python program that processes multiple products and quantities, applies category-specific taxes, calculates threshold-based discounts, enforces a maximum concession limit, and generates an itemized invoice.

2. Objectives

• Accept and process multiple product records.

• Calculate each line amount from unit price × quantity.

• Apply a threshold discount to the pre-tax subtotal.

• Cap the discount at the maximum concession of Rs. 2,000.

• Apply different tax rates by product category and print an itemized invoice.

3. Assumptions and Business Rules

Assumptions

1. The customer can purchase multiple products in a single transaction.
2. Each product has a unique product name or ID, unit price, and category.
3. The quantity of each product must be a positive integer.
4. Each product category has a predefined tax rate.
5. Discounts are applied based on predefined purchase amount thresholds.
6. A maximum concession limit is specified to restrict the total discount.
7. All product prices and billing amounts are calculated in Indian Rupees (₹).
8. The customer receives an itemized invoice containing product details, discounts, taxes, and the final payable amount.
9. Product prices and tax rates are assumed to be valid and non-negative.
10. The billing system processes one customer transaction at a time.

Business Rules

1. Product Selection: The system must accept multiple products with their respective prices, categories, and quantities.

2. Subtotal Calculation: Calculate each product's subtotal using:

```
Subtotal = Unit Price × Quantity
```

3. Category-Specific Tax: Apply the appropriate tax rate based on each product's category.

```
Tax Amount = Taxable Amount × Tax Rate / 100
```

4. Threshold Discount: Apply discounts when the purchase subtotal reaches the predefined minimum purchase amount.

5. Multiple Discount Rules: Evaluate the applicable discount rules and select or combine discounts according to the configured discount policy.

6. Maximum Concession Limit: The total discount must not exceed the specified maximum concession amount.

7. Discount Validation: The discount amount must not be negative or greater than the eligible purchase amount.

8. Final Bill Calculation: Calculate the final payable amount after applying discounts and adding applicable taxes.

```
Final Bill = Subtotal – Discount + Tax
```

9. Invoice Generation: Display product name, quantity, unit price, subtotal, discount, tax, and final amount in an itemized invoice.

10. Invalid Input Handling: Reject zero or negative quantities, invalid prices, and invalid tax rates.

11. Rounding Rule: Round monetary amounts to two decimal places.

12. Total Verification: Ensure that the invoice total matches the sum of all product amounts, discounts, and taxes.

4. Algorithm

1. Start the program. Rs. 2,000

2. Store the product details: name, category, quantity, and unit price.

3. Calculate the pre-tax subtotal from all line amounts.

4. Select the discount rate using the subtotal thresholds.

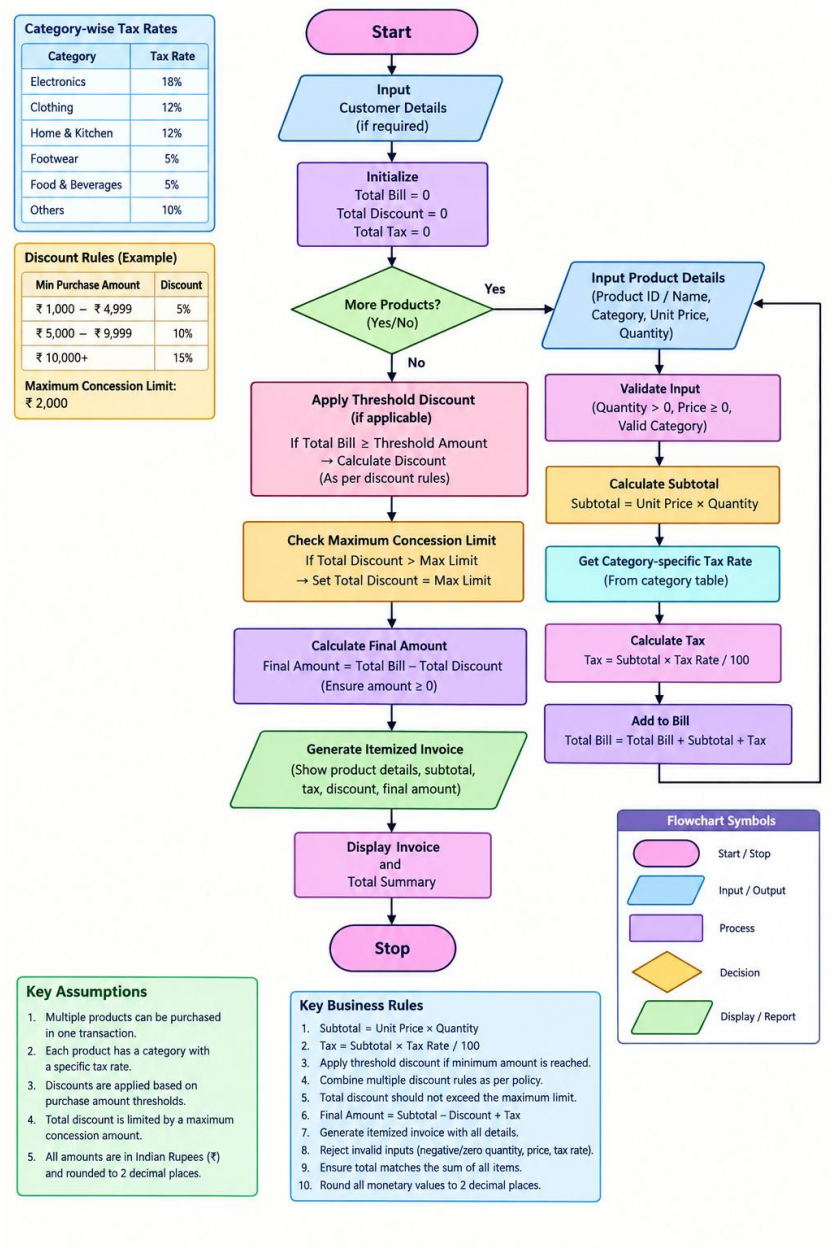
5. Calculate discount = minimum(subtotal × rate, maximum concession).

6. Allocate the discount proportionally to each product line.

7. Calculate category tax on each discounted line amount.

8. Add discounted amounts and taxes to obtain the grand total. 9. Print the itemized invoice and stop.

5. Flowchart



6. Pseudocode

BEGIN

INPUT numberOfProducts

INPUT maximumConcessionLimit

```
SET totalSubtotal = 0
SET totalTax = 0
SET totalDiscount = 0

FOR i = 1 TO numberOfProducts

  INPUT productName
  INPUT category
  INPUT unitPrice
  INPUT quantity

  IF unitPrice < 0 OR quantity <= 0 THEN
    PRINT "Invalid Product Details"
    STOP
  END IF

  SET subtotal = unitPrice * quantity

  // Determine category-specific tax rate
  IF category = "Electronics" THEN
    SET taxRate = 18
  ELSE IF category = "Clothing" THEN
    SET taxRate = 12
  ELSE IF category = "Home & Kitchen" THEN
    SET taxRate = 12
```

```
    ELSE IF category = "Footwear" THEN
        SET taxRate = 5
    ELSE IF category = "Food & Beverages" THEN
        SET taxRate = 5
    ELSE IF category = "Others" THEN
        SET taxRate = 10
    ELSE
        PRINT "Invalid Category"
        STOP
    END IF

    SET tax = subtotal * taxRate / 100

    SET totalSubtotal = totalSubtotal + subtotal
    SET totalTax = totalTax + tax

    STORE productName, category, quantity,
          unitPrice, subtotal, taxRate, tax

END FOR

// Apply threshold discount
IF totalSubtotal >= 10000 THEN
    SET discountRate = 15
ELSE IF totalSubtotal >= 5000 THEN
    SET discountRate = 10
```

```
ELSE IF totalSubtotal >= 1000 THEN
   SET discountRate = 5
ELSE
   SET discountRate = 0
END IF

SET totalDiscount = totalSubtotal * discountRate / 100

// Apply maximum concession limit
IF totalDiscount > maximumConcessionLimit THEN
   SET totalDiscount = maximumConcessionLimit
END IF

// Calculate final bill
SET finalAmount = totalSubtotal + totalTax - totalDiscount

// Generate itemized invoice
PRINT "========== RETAIL INVOICE =========="

FOR each product IN storedProducts
   PRINT productName
   PRINT category
   PRINT quantity
   PRINT unitPrice
   PRINT subtotal
   PRINT taxRate
```

```
    PRINT tax
  END FOR

  PRINT "Total Subtotal: ", totalSubtotal
  PRINT "Total Tax: ", totalTax
  PRINT "Total Discount: ", totalDiscount
  PRINT "Maximum Concession Limit: ", maximumConcessionLimit
  PRINT "Final Payable Amount: ", finalAmount

  PRINT "==================================="

END
```

7. Python Source Code

```
# Retail Billing with Multiple Discount Rules

# Category-wise tax rates
tax_rates = {
  "electronics": 18,
  "clothing": 12,
  "home & kitchen": 12,
  "footwear": 5,
  "food & beverages": 5,
  "others": 10
}
```

```
# Discount rules
MAX_CONCESSION = 2000

products = []
total_subtotal = 0
total_tax = 0

print("===== RETAIL BILLING SYSTEM =====")

try:
    n = int(input("Enter number of products: "))

    if n <= 0:
        print("Invalid number of products")

    else:
        for i in range(n):
            print(f"\nProduct {i + 1}")

            name = input("Enter product name: ").strip()
            category = input("Enter category: ").strip().lower()
            price = float(input("Enter unit price: "))
            quantity = int(input("Enter quantity: "))

            # Validate product details
            if not name or category not in tax_rates:
```

```
    print("Invalid product name or category")
    raise ValueError

if price < 0 or quantity <= 0:
    print("Invalid price or quantity")
    raise ValueError

# Calculate subtotal
subtotal = price * quantity

# Calculate category-specific tax
tax_rate = tax_rates[category]
tax = subtotal * tax_rate / 100

total_subtotal += subtotal
total_tax += tax

products.append({
    "name": name,
    "category": category.title(),
    "price": price,
    "quantity": quantity,
    "subtotal": subtotal,
    "tax_rate": tax_rate,
    "tax": tax
})
```

```
# Apply threshold discount
if total_subtotal >= 10000:
    discount_rate = 15
elif total_subtotal >= 5000:
    discount_rate = 10
elif total_subtotal >= 1000:
    discount_rate = 5
else:
    discount_rate = 0

discount = total_subtotal * discount_rate / 100

# Apply maximum concession limit
discount = min(discount, MAX_CONCESSION)

# Calculate final amount
final_amount = total_subtotal + total_tax - discount

# Display itemized invoice
print("\n" + "=" * 75)
print("                RETAIL INVOICE")
print("=" * 75)

print(
    f"{'Product':<15}{'Category':<18}"
```

```
    f"{'Qty':>5}{'Price':>10}{'Subtotal':>12}"
)
print("-" * 75)

for p in products:
    print(
        f"{p['name'][:14]:<15}"
        f"{p['category'][:17]:<18}"
        f"{p['quantity']:>5}"
        f"{p['price']:>10.2f}"
        f"{p['subtotal']:>12.2f}"
    )
    print(
        f"   Tax Rate: {p['tax_rate']}% | "
        f"Tax Amount: Rs. {p['tax']:.2f}"
    )

print("-" * 75)
print(f"Total Subtotal      : Rs. {total_subtotal:.2f}")
print(f"Total Tax           : Rs. {total_tax:.2f}")
print(f"Discount Rate       : {discount_rate}%")
print(f"Total Discount      : Rs. {discount:.2f}")
print(f"Maximum Concession   : Rs. {MAX_CONCESSION:.2f}")
print(f"FINAL PAYABLE AMOUNT : Rs. {final_amount:.2f}")
print("=" * 75)
print("Thank you for shopping!")
```

except ValueError:

print("Please enter valid product details and numeric values.")

8. Output Screenshot

```
Enter number of products: 7

Product 1
Enter product name:
============== RESTART: C:/Users/LEKHANA/OneDrive/Desktop/activity 4.py ==============
===== RETAIL BILLING SYSTEM =====
Enter number of products:
============== RESTART: C:/Users/LEKHANA/OneDrive/Desktop/activity 4.py ==============
===== RETAIL BILLING SYSTEM =====
Enter number of products: 2

Product 1
Enter product name: Laptop
Enter category: electronics
Enter unit price: 40000
Enter quantity: 1

Product 2
Enter product name: Shirt
Enter category: clothing
Enter unit price: 1000
Enter quantity: 2

================================================================================
                    RETAIL INVOICE
================================================================================
Product        Category           Qty     Price    Subtotal
--------------------------------------------------------------------------------
Laptop         Electronics          1  40000.00    40000.00
   Tax Rate: 18% | Tax Amount: Rs. 7200.00
Shirt          Clothing             2   1000.00     2000.00
   Tax Rate: 12% | Tax Amount: Rs. 240.00
--------------------------------------------------------------------------------
Total Subtotal       : Rs. 42000.00
Total Tax            : Rs. 7440.00
Discount Rate        : 15%
Total Discount       : Rs. 2000.00
Maximum Concession   : Rs. 2000.00
FINAL PAYABLE AMOUNT : Rs. 47440.00
================================================================================
Thank you for shopping!
```

9. Test Cases

1. **TC01 – Single Product Billing:** Verify that the bill is generated correctly for a single product with a valid price and quantity.

2. **TC02 – Multiple Products:** Check whether the system calculates the subtotal correctly for multiple products in one transaction.

3. **TC03 – Quantity Calculation:** Verify that the product subtotal is calculated as unit price multiplied by quantity.

4. **TC04 – Category-Specific Tax:** Check whether the correct tax rate is applied according to each product category.

5. **TC05 – Minimum Discount Threshold:** Verify that a 5% discount is applied when the total subtotal reaches ₹1,000.

6. **TC06 – Medium Discount Threshold:** Check whether a 10% discount is applied when the total subtotal reaches ₹5,000.

7. **TC07 – Highest Discount Threshold:** Verify that a 15% discount is applied when the total subtotal reaches ₹10,000 or more.

8. **TC08 – Maximum Concession Limit:** Check whether the total discount is restricted to the maximum concession limit of ₹2,000.

9. **TC09 – No Discount:** Verify that no discount is applied when the total subtotal is below ₹1,000.

10. **TC10 – Final Bill Calculation:** Check whether the final payable amount is calculated a total subtotal + total tax − total discount.

11. **TC11 – Invalid Product Category:** Verify that the system rejects an unsupported product category.

12. **TC12 – Invalid Price:** Check whether negative product prices are rejected.

13. **TC13 – Invalid Quantity:** Verify that zero or negative quantities are rejected.

14. **TC14 – Invalid Number of Products:** Check whether the system rejects zero or negative product counts.

15. **TC15 – Itemized Invoice:** Verify that the invoice displays product names, categories, quantities, unit prices, subtotals, tax rates, tax amounts, discounts, and the final payable amount.

16. **TC16 – Decimal Price Handling:** Check whether decimal prices and monetary calculations are displayed to two decimal places.

17. **TC17 – Input Validation:** Verify that non-numeric values entered for price, quantity, or product count are handled appropriately.

18. **TC18 – Total Verification:** Check whether the total subtotal, tax, discount, and final amount match the individual product calculations.

10. Conclusion

The program demonstrates multi-product billing, quantity-based line totals, category-specific taxes, subtotal threshold discounts, a maximum concession cap, and itemized invoice generation. Tax percentages and discount policies are sample assumptions; replace them with the rules specified by your faculty or retailer if they differ.